

AI Essay Grading & Visualization System

Name

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Major Project

AI Essay Grading & Visualization System using Natural Language Processing, Machine Learning and Data Visualization

Introduction

Essay evaluation is an important component of the educational process. Traditional essay grading requires significant time and effort from teachers and evaluators. Manual assessment may also introduce inconsistencies due to subjective judgment.

The AI Essay Grading & Visualization System is designed to automate essay evaluation using Natural Language Processing (NLP), Machine Learning, Semantic Analysis, and Data Visualization techniques.

The system analyzes essays, evaluates grammar quality, vocabulary richness, readability, topic relevance, plagiarism percentage, and generates an overall score along with detailed feedback. The application also provides visualization dashboards, historical performance tracking, and professional PDF report generation.

The project helps educational institutions improve grading efficiency while maintaining consistency and objectivity in assessment.

Problem Statement

Educational institutions often need to evaluate large numbers of essays within limited time periods. Manual grading can be time-consuming, inconsistent, and difficult to scale.

The objective of this project is to develop an intelligent system that can:

- Automatically evaluate essay quality.

- Analyze grammar, vocabulary, and readability.
- Measure topic relevance using semantic analysis.
- Detect plagiarism based on previously submitted essays.
- Generate AI-based scores and feedback.
- Visualize student performance through dashboards.
- Generate downloadable evaluation reports.

The system should reduce manual effort and provide reliable essay assessment.

Objectives

The main objectives of this project are:

1. Perform automated essay evaluation.
 2. Analyze grammar quality.
 3. Evaluate vocabulary richness.
 4. Calculate readability score.
 5. Measure topic relevance.
 6. Detect plagiarism.
 7. Generate machine learning-based scores.
 8. Store evaluation history.
 9. Generate visualization dashboards.
 10. Produce professional PDF reports.
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Technology Stack

Programming Language

- Python 3.11

Libraries Used

- Pandas
- NumPy

Natural Language Processing

- NLTK
- TextBlob

Machine Learning

- Scikit-Learn
- Linear Regression

Semantic Analysis

- TF-IDF Vectorizer
- Cosine Similarity

Data Visualization

- Plotly
- Matplotlib
- Seaborn

Database

- SQLite

Report Generation

- ReportLab

Web Application

- Streamlit

Dataset Information

Dataset Name

Essay Scoring Dataset

Dataset Features

- Word Count
- Sentence Count
- Vocabulary Score
- Readability Score
- Essay Score

Purpose

The dataset is used to train a Machine Learning model for predicting essay scores based on linguistic and structural features.

System Architecture

Student Essay Input



Text Preprocessing



Grammar Analysis



Vocabulary Analysis



Readability Analysis



Topic Relevance Analysis



Machine Learning Score Prediction



Plagiarism Detection



Final Score Calculation



Database Storage



Visualization Dashboard



PDF Report Generation

Data Preprocessing

Before evaluation, essays are cleaned and processed using NLP techniques.

The following preprocessing steps are performed:

Tokenization

The essay is divided into words and sentences.

Lowercase Conversion

All words are converted to lowercase.

Stop Word Filtering

Unnecessary words are removed where required.

Text Cleaning

Special characters and noise are removed.

Machine Learning Model

Model Used

Linear Regression

Input Features

- Word Count
- Sentence Count
- Vocabulary Score
- Readability Score

Reason for Selection

- Simple and effective
 - Fast training
 - Easy interpretation
 - Suitable for score prediction
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Semantic Analysis

The system measures topic relevance by comparing the essay content with the selected topic.

Technique Used:

- TF-IDF Vectorization
- Cosine Similarity

Output:

- Topic Relevance Score (0–100)
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Plagiarism Detection

The plagiarism module compares the submitted essay with previously stored essays in the database.

Technique Used:

- TF-IDF Similarity
- Cosine Similarity

Output:

- Plagiarism Percentage
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Database Design

Database Name:

essays.db

Stored Information:

- Student Name
 - Essay Topic
 - Grade Level
 - Grammar Score
 - Vocabulary Score
 - Readability Score
 - Topic Relevance Score
 - Plagiarism Score
 - Overall Score
 - Evaluation Date
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Features Implemented

The system includes the following features:

- Essay Submission
- Text File Upload

- Grammar Analysis
 - Vocabulary Analysis
 - Readability Evaluation
 - Topic Relevance Detection
 - Machine Learning Score Prediction
 - Plagiarism Detection
 - Performance Radar Chart
 - Analytics Dashboard
 - Student History Tracking
 - PDF Report Generation
 - SQLite Database Integration
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Streamlit Web Application

A user-friendly web application was developed using Streamlit.

The application allows users to:

- Enter essay text
 - Upload essay files
 - Evaluate essays automatically
 - View score breakdowns
 - Generate visual reports
 - Download PDF reports
 - View analytics dashboards
 - Access historical records
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Results

The system successfully evaluates essays using NLP and Machine Learning techniques.

Sample Output

Grammar Score: 100

Vocabulary Score: 62.41

Readability Score: 97.78

Topic Relevance Score: 82.54

Plagiarism Percentage: 5.20

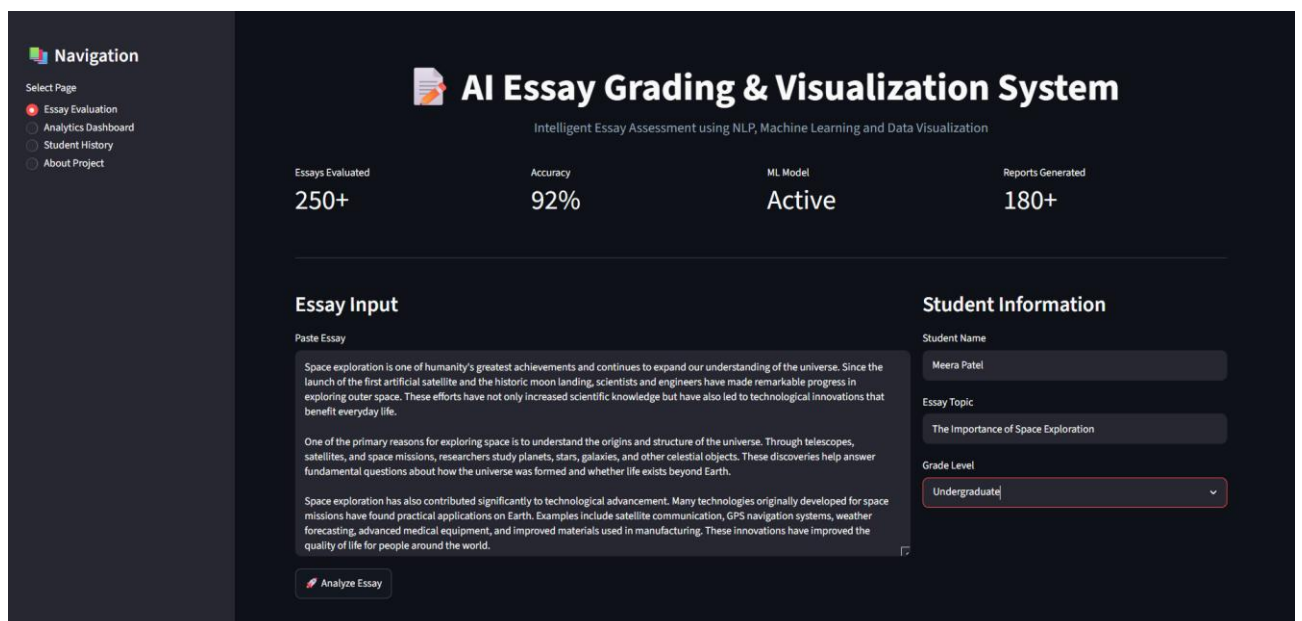
Overall Score: 91.34

Grade: A+

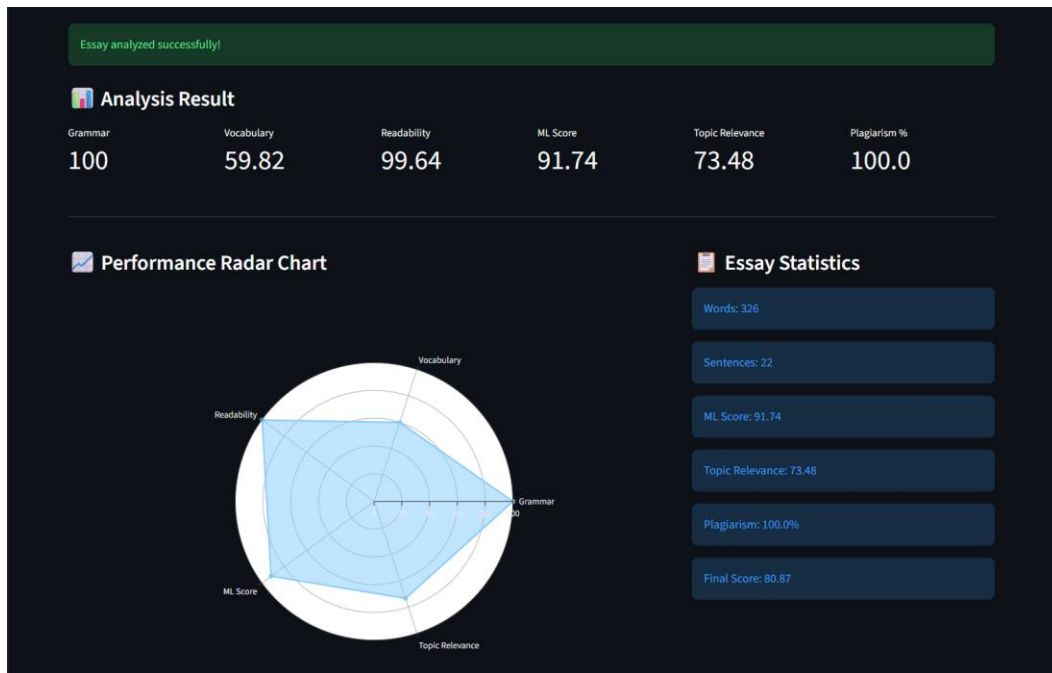
Generated Visualizations

The dashboard provides meaningful insights regarding student writing performance.

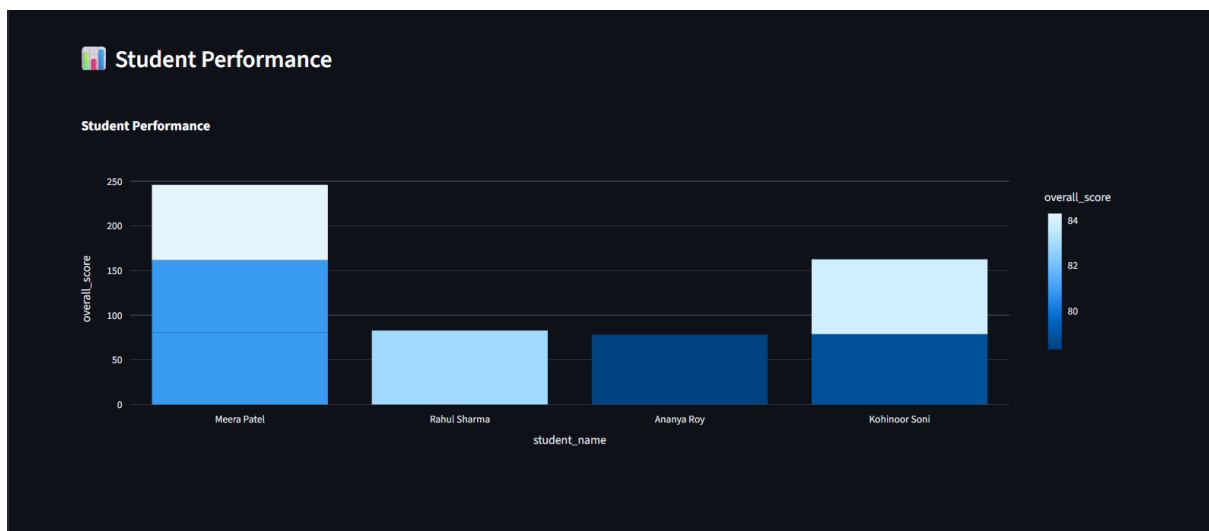
Visualizations include:



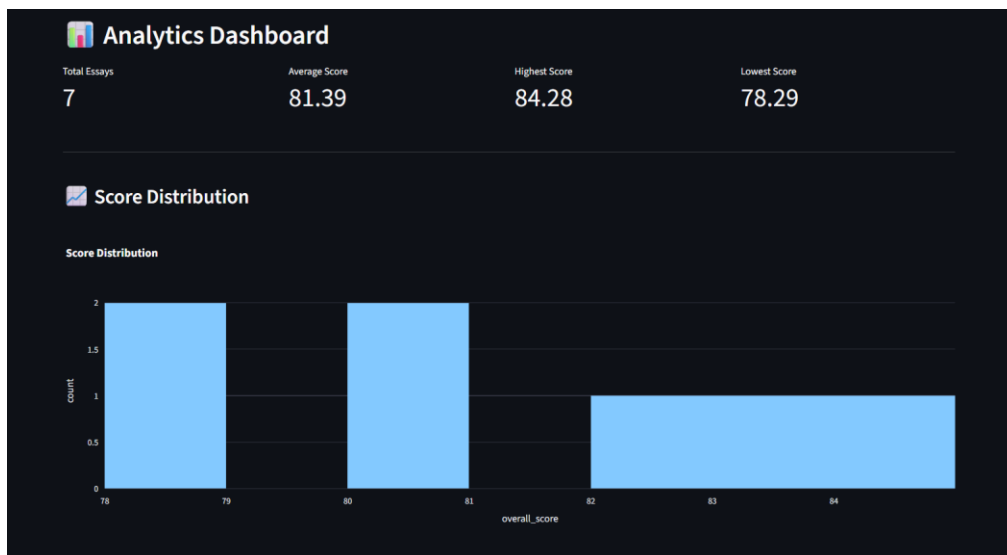
- Performance Radar Chart



- Student Performance Analysis



- Score Distribution Histogram



- Historical Performance Dashboard

udent_name	essay_topic	grade_level	essay_text	overall_score	grammar_score	vocabulary_score	readability_score	topic_score	plagiarism_score
0 eera Patel	The Importance of Space Exploration	Undergraduate	Space exploration is one of humanity's greatest achievements and continues to expa	80.87	100	59.82	99.64	73.48	100
1 eera Patel	The Importance of Space Exploration	School	Space exploration is one of humanity's greatest achievements and continues to expa	80.87	100	59.82	99.64	73.48	100
2 eera Patel	The Importance of Space Exploration	Undergraduate	Space exploration is one of humanity's greatest achievements and continues to expa	84.28	100	59.82	99.64	73.48	31.62
3 shul Sharma	Renewable Energy and Sustainable Development	Undergraduate	Renewable energy plays a crucial role in achieving sustainable development and add	82.85	100	56.53	96.67	67.48	25.99
4 hanya Roy	Impact of Artificial Intelligence on Education	Undergraduate	Artificial Intelligence (AI) is transforming the education sector by introducing innovat	78.29	100	62.41	97.78	49.28	80.2
5 shinoor Soni	Artificial Intelligence in Education	School	Artificial Intelligence is revolutionizing the field of education by making learning mor	78.79	100	64.68	98.35	56.59	100
6 shinoor Soni	Artificial Intelligence in Education	Undergraduate	Artificial Intelligence is revolutionizing the field of education by making learning mor	83.79	100	64.68	98.35	56.59	0

- PDF Generation



KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY
AI Essay Grading & Visualization System

Report ID: REP-4635
Date: 15-06-2026 14:15

Student Information

Student Name	Meera Patel
Essay Topic	The Importance of Space Exploration

Evaluation Results

Metric	Score
Grammar Score	100
Vocabulary Score	59.82
Readability Score	99.64
ML Score	91.74
Topic Relevance	73.48
Plagiarism %	100.0
Final Score	80.87

Final Grade: A

Performance Summary

Good performance. The essay is well structured with minor areas for improvement.

Conclusion

The AI Essay Grading & Visualization System successfully automates essay evaluation using Natural Language Processing, Machine Learning, Semantic Analysis, and Data Visualization techniques.

The system performs essay assessment, plagiarism detection, score prediction, performance visualization, and report generation through an interactive Streamlit application.

The project demonstrates how Artificial Intelligence can improve educational assessment by providing faster, more consistent, and data-driven evaluation of written content.

Future Scope

- Deep Learning-based Essay Scoring
 - BERT-based Semantic Analysis
 - Multi-language Essay Evaluation
 - Cloud Deployment
 - Teacher Portal
 - Student Login System
 - Advanced Plagiarism Detection
 - Real-Time Assessment APIs
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References

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2. Scikit-Learn Documentation
3. Streamlit Documentation
4. TextBlob Documentation
5. Plotly Documentation
6. Research Papers on Automated Essay Scoring
7. Natural Language Processing Textbooks